SALAS O'BRIEN
expect a difference

SALAS O'BRIEN
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Project:	**Traditions of Braselton; Sterling Residential**	Date:	February 17, 2021
Location:	Jefferson, GA	By:	RCS / TJH
Description:	Dogwood - Slab	House Faces:	Worst East

These Loads are dependent on the information/assumptions listed below. **Any deviation from these assumptions during construction could have a significant impact on the cooling and heating loads for this house and render the recommended equipment sizing invalid.** If any of these assumptions are incorrect, please let us know immediately so we can re-calculate them. This information will be used in a residential heating and cooling load program based on calculations recommended by ASHRAE, Handbook of Fundamentals. Equipment sizing indicated in this report is valid only upon verification that actual construction is in full compliance with the descriptions below and independent testing of HVAC systems has been performed; such verification and testing shall be performed by Vallus.

BUILDING DATA		
Typical Ceiling Height	9.0	ft
Floor Area	2,674	ft^2
Volume	25,596	ft^3
Percentage Glass	8	%
Number of Bedrooms	4	

DESIGN CONDITIONS	Summer	Winter
Outdoor	95 °Fdb	18 °Fdb
Indoor	75 °Fdb	72 °Fdb
ΔT	20 °Fdb	54 °Fdb
Range	17.3 °Fdb	

ENERGY CALCULATIONS	
Right Choice HVAC kWh	10343
SEER	14
ACH50	5.0
Mechancial Ventilation	NO
House Outside Air CFM	N/A

CONSTRUCTION DESCRIPTIONS

	#	U-Value	Description/Notes		
Walls / Partitions	W1	0.077	Above Grade	2x4	R-13 batt
	W2	0.067	Basement	Concrete + 2x4	R-13 batt
	W3	0.053	Attic	2x6	R-19 batt
Doors	**#**	**U-Value**	**Description/Notes**		
	D1	0.5	Exterior Door		R-2
	D2	0.5	Garage Door		R-2
Roofs	**#**	**U-Value**	**Description/Notes**		
	R1	0.031	Flat Ceiling	2x6	R-30 sprayed
	R2	0.033	Vaulted	2x8	R-30 sprayed
Floors	**#**	**U-Value**	**Description/Notes**		
	F1	0.053	Framed		R-19 batt
	F2	0.1	Slab		None

Glass	#	Glass Type	Windows	U-Value	Internal Shading	Notes
	G1	Double Insulated	All types	0.33	Blinds/Draperies	SHGC 0.19
	G2					
	G3					

LOAD SUMMARY

			Unit 1 - First Floor		Unit 2 - Second Floor	
			Cooling	**Heating**	**Cooling**	**Heating**
Sensible Load			13319 Btu/hr	17811 Btu/hr	14533 Btu/hr	17803 Btu/hr
Minimum Nominal Capacity			1.48 Tons	5.22 kW	1.61 Tons	5.22 kW
Load Density			799 SF/Ton	4.4 W/SF	924 SF/Ton	3.5 W/SF
Floor Area Served			1182.0 SF	1182.0 SF	1492.0 SF	1492.0 SF
Mechanical Ventilation			N/A CFM	N/A CFM	N/A CFM	N/A CFM
System Size			1.5 Tons	8.0 kW	2.0 Tons	8.0 kW
Leaving Air Temperature			57.2 deg F	112.1 deg F	57.2 deg F	101.6 deg F
Installed Capacity			788 SF/Ton	6.8 W/SF	746 SF/Ton	5.4 W/SF

I understand the information provided herein, and warrant, to the best of my ability and knowledge, and to the full extent of my control as the Contractor, that this house will be built in accordance with the information given above.

Signed________________________ Title________________ Date_________

Project:	**Traditions of Braselton; Sterling Residential**	Date:	February 17, 2021
Location:	Jefferson, GA	By:	RCS / TJH
Description:	Dogwood - Slab	House Faces: Worst	East

Unit 1	**First Floor**			
Size	**1.50**	**Tons**		
Airflow	**600**	**CFM**		
Heat kW	**8**	**kW**		
Room	**CFM Cool**	**CFM Heat**	**CFM Avg**	**Duct Size**
Foyer	**72**	**121**	**97**	
Mud Rm	**32**	**40**	**36**	
Powder	**13**	**12**	**13**	
Utility	**54**	**27**	**41**	
Great Rm	**183**	**153**	**168**	
Breakfast	**37**	**66**	**52**	
Kitchen	**151**	**68**	**110**	
Pantry	**11**	**24**	**18**	
Dining	**47**	**87**	**67**	
-	-	-	-	
-	-	-	-	
-	-	-	-	
-	-	-	-	
-	-	-	-	

Unit 2	**Second Floor**			
Size	**2.00**	**Tons**		
Airflow	**800**	**CFM**		
Heat kW	**8**	**kW**		
Room	**CFM Cool**	**CFM Heat**	**CFM Avg**	**Duct Size**
Stairs	**47**	**29**	**38**	
Landing	**55**	**19**	**37**	
Master Bed	**151**	**165**	**158**	
Master Sitting	**49**	**80**	**65**	
Master WIC (Left)	**64**	**72**	**68**	
Master WIC (Interior)	**9**	**7**	**8**	
Master Bath	**107**	**119**	**113**	
Bed 2	**95**	**72**	**83**	
Bed 4	**109**	**107**	**108**	
Bath	**19**	**23**	**21**	
Bed 3	**94**	**106**	**100**	
-	-	-	-	
-	-	-	-	
-	-	-	-	

Project:	**Traditions of Braselton; Sterling Residential**	Date:	February 17, 2021	Typical Ceiling Height	9
Location:	Jefferson, GA	By:	RCS / TJH		
Description:	Dogwood - Slab	Level(s):	First Floor	Kneewall Height	1
	Square Feet 1182 Volume 11718 SF/Ton 788	SEER	14		

Cooling Size	1.5	DX Cooling kWh	1776
Heating Size	1.5	RC Heating kWh	3315
Best Case	1.5	Total kWh	5091

COOLING LOAD CALCULATION

					Ceiling Height			19		9		9		9		9		9		9		9		9		9		9		9		9		9	
Components			CLTD (°F)	U-Value (Btu/hr-sf-°F)	Cooling Load Factor	Entire House		Foyer		Mud Rm		Powder		Utility		Great Rm		Breakfast		Kitchen		Pantry		Dining											
	Type	Description				Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr
Glass	G1	North	Right	(E)	6 Btu/hr-sf	9	51		-		-		-		-		-		-	9	51		-		-		-		-		-		-		-
	G1	NorthEast		(SE)	7 Btu/hr-sf	0	0		-		-		-		-		-		-		-		-		-		-		-		-		-		-
	G1	East	Front	(S)	12 Btu/hr-sf	30	348	15	174		-		-		-		-		-		-		-	15	174		-		-		-		-		-
	G1	SouthEast		(SW)	13 Btu/hr-sf	0	0		-		-		-		-		-		-		-		-		-		-		-		-		-		-
	G1	South	Left	(W)	14 Btu/hr-sf	0	0		-		-		-		-		-		-		-		-		-		-		-		-		-		-
	G1	SouthWest		(NW)	23 Btu/hr-sf	0	0		-		-		-		-		-		-		-		-		-		-		-		-		-		-
	G1	West	Back	(N)	26 Btu/hr-sf	45	1161		-		-		-		-	45	1,161		-		-		-		-		-		-		-		-		-
	G1	NorthWest		(NE)	16 Btu/hr-sf	0	0		-		-		-		-		-		-		-		-		-		-		-		-		-		-
	G1	Sky Light			40 Btu/hr-sf	0	0		-		-		-		-		-		-		-		-		-		-		-		-		-		-
	G1	Shaded			6 Btu/hr-sf	52	298	8.938	51		-		-		-		-	28.36	161		-		-	15	85		-		-		-		-		-
Door	D1	Door	21	0.50	10.50 Btu/hr-sf	24	257	20	210		-		-		-		-	4.452	47		-		-		-		-		-		-		-		-
	D2	Door	30	0.50	15.00 Btu/hr-sf	18	267		-	17.81	267		-		-		-		-		-		-		-		-		-		-		-		-
Above Grade Walls	W1	North	13	0.08	1.00 Btu/hr-sf	450	451	133	134	-	-	-	-	-	-	-	-	-	-	108	108	68	68	141	141	-	-	-	-	-	-	-	-	-	-
	W1	NorthEast	19	0.08	1.46 Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
	W1	East	23	0.08	1.77 Btu/hr-sf	207	367	124	220	-	-	-	-	-	-	-	-	-	-	-	-	-	-	83	146	-	-	-	-	-	-	-	-	-	-
	W1	SouthEast	21	0.08	1.62 Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
	W1	South	16	0.08	1.23 Btu/hr-sf	237	292	9	12	-	-	-	-	63	78	164	202	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
	W1	SouthWest	21	0.08	1.62 Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
	W1	West	23	0.08	1.77 Btu/hr-sf	336	596	-	-	-	-	-	-	-	-	173	307	50	88	66	117	48	84	-	-	-	-	-	-	-	-	-	-	-	-
	W1	NorthWest	19	0.08	1.46 Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Basement Walls	W1	North	13	0.08	1.00 Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
	W1	NorthEast	19	0.08	1.46 Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
	W1	East	23	0.08	1.77 Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
	W1	SouthEast	21	0.08	1.62 Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
	W1	South	16	0.08	1.23 Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
	W1	SouthWest	21	0.08	1.62 Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
	W1	West	23	0.08	1.77 Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
	W1	NorthWest	19	0.08	1.46 Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Other	W2	Grade	0	0.07	0.00 Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
	W1	Partition	10	0.08	0.77 Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
	W1	Garage	30	0.08	2.31 Btu/hr-sf	287	663	48	111	71	164	84	194	84	195	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
	W3	Attic	15	0.05	0.80 Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
	W3	Kneewall	15	0.05	0.80 Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Roof	R1	Ceiling	55	0.03	1.71 Btu/hr-sf	99	169	99	169		-		-		-		-		-		-		-		-		-		-		-		-		-
	R2	Vaulted	55	0.03	1.82 Btu/hr-sf	0	0		-		-		-		-		-		-		-		-		-		-		-		-		-		-
Floor	F1	Crawl/Uncon.	11	0.05	0.58 Btu/hr-sf	0	0		-		-		-		-		-		-		-		-		-		-		-		-		-		-
	F1	Garage	30	0.05	1.59 Btu/hr-sf	0	0		-		-		-		-		-		-		-		-		-		-		-		-		-		-
	F2	Slab	0	0.10	0.00 Btu/hr-sf	1182	0	108	-	83	-	30	-	53	-	359	-	179	-	180	-	30	-	160	-		-		-		-		-		-
Internal	People				255 Btu/hr	1	255		-		-		-		-	1	255		-		-		-		-		-		-		-		-		-
	Lights (W)				3.413 Btu/hr-W	593	2024	54	184	42	143	15	51	27	92	180	614	90	307	90	307	15	51	80	273	-	-	-	-	-	-	-	-	-	-
	Appliances (W)				3.413 Btu/hr-W	1130	3857		-		-		-	200	683	250	853		-	680	2,321		-		-		-		-		-		-		-
Vent	Infiltration		0.25	ACH	0.09 Btu/hr-cf		1055		185		67		24		43		291		145		146		24		130		-		-		-		-		-
	Volume				CF	11718		2052		747		270		477		3231		1611		1620		270		1440		0		0		0		0		0	
	Area				SF	1182		108		83		30		53		359		179		180		30		160											
Subtotal (Btu/hr)							12108		1449		641		270		1090		3683		748		3049		228		949		0		0		0		0		0
Safety Factor				10 %			1211																												
Sensible Cooling Load (Btu/hr)							**13319**																												
Nominal Tons Required (75% sens. cap.)							**1.48**																												
System Selection				**1.5 Tons**	**Space CFM**		**600**		**72**		**32**		**13**		**54**		**183**		**37**		**151**		**11**		**47**		**-**		**-**		**-**		**-**		**-**

HEATING LOAD CALCULATION

Components			ΔT (°F)	U-Value	Heating Load Factor	Entire House		Foyer		Mud Rm		Powder		Utility		Great Rm		Breakfast		Kitchen		Pantry		Dining		0		0		0		0		0	
	Type	Description				Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr
Glass	G1		54	0.33	17.8 Btu/hr-sf	136	2429	23.94	427	0	-	0	-	0	-	45	802	28.36	505	9	160	0	-	30	535	0	-	0	-	0	-	0	-	0	-
	G2		54	0.00	0.0 Btu/hr-sf	0	0	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-
	G3		54	0.00	0.0 Btu/hr-sf	0	0	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-
Door	D1		54	0.50	27.0 Btu/hr-sf	24	660	20	540	0	-	0	-	0	-	0	-	4.452	120	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-
	D2		54	0.50	27.0 Btu/hr-sf	18	481	0	-	17.81	481	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-
Walls	W1		54	0.08	4.2 Btu/hr-sf	1230	5116	267.3	1,112	0	-	0	-	63.3	263	337.5	1,403	49.69	207	173.5	721	115.8	481	223.3	928	0	-	0	-	0	-	0	-	0	-
	W2		54	0.07	3.6 Btu/hr-sf	0	0	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-
	W3		54	0.05	2.9 Btu/hr-sf	0	0	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-
	W1	Partition	27	0.08	2.1 Btu/hr-sf	287	597	48.13	100	70.8	147	84.1	175	84.2	175	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-
	W2	Grade	27	0.07	1.8 Btu/hr-sf	0	0	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-
Roof	R1		54	0.03	1.7 Btu/hr-sf	99	166	99	166	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-
	R2		54	0.03	1.8 Btu/hr-sf	0	0	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-
Floor	F1		54	0.05	2.9 Btu/hr-sf	0	0	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-
	F2	Slab	27	0.10	2.7 Btu/hr-sf	1182	3191	108	292	83	224	30	81	53	143	359	969	179	483	180	486	30	81	160	432	0	-	0	-	0	-	0	-	0	-
Area					SF	1182		108		83		30		53		359		179		180		30		160		0		0		0		0		0	
Infiltration			0.25	ACH	0.24 Btu/hr	0	2847		499		182		66		116		785		391		394		66		350		-		-		-		-		-
Subtotal (Btu/hr)							15488		3134		1034		321		697		3960		1707		1761		628		2245		0		0		0		0		0
Safety Factor				15 %			2323																												
Heat Loss (Btu/hr)							**17811**																												
Minimum Heater Size (kW)							**5.22**																												
System Selection				**8 kW**	**Htg. CFM**		**600**		**121**		**40**		**12**		**27**		**153**		**66**		**68**		**24**		**87**		**-**		**-**		**-**		**-**		**-**

Project:	**Traditions of Braselton; Sterling Residential**	Date:	February 17, 2021	Typical Ceiling Height	9
Location:	Jefferson, GA	By:	RCS / TJH		
Description:	Dogwood - Slab	Level(s):	Second Floor	Kneewall Height	1
	Square Feet 1492 Volume 13878 SF/Ton 746	SEER	14		

Cooling Size	2	DX Cooling kWh	1938
Heating Size	1.5	RC Heating kWh	3314
Best Case	1.5	Total kWh	5251

COOLING LOAD CALCULATION

Components	Type	Description	CLTD (°F)	U-Value (Btu/hr-sf-°F)	Cooling Load Factor	Entire House Qty	Entire House Btu/hr	Stairs Qty	Stairs Btu/hr	Landing Qty	Landing Btu/hr	Master Bed Qty	Master Bed Btu/hr
					Ceiling Height			19		9		9	
Glass	G1	North	Right	(E)	6 Btu/hr-sf	0	0		-		-		-
Glass	G1	NorthEast		(SE)	7 Btu/hr-sf	0	0		-		-		-
Glass	G1	East	Front	(S)	12 Btu/hr-sf	45	522		-		-		-
Glass	G1	SouthEast		(SW)	13 Btu/hr-sf	0	0		-		-		-
Glass	G1	South	Left	(W)	14 Btu/hr-sf	30	419		-		-	30	419
Glass	G1	SouthWest		(NW)	23 Btu/hr-sf	0	0		-		-		-
Glass	G1	West	Back	(N)	26 Btu/hr-sf	46	1187		-		-		-
Glass	G1	NorthWest		(NE)	16 Btu/hr-sf	0	0		-		-		-
Glass	G1	Sky Light			40 Btu/hr-sf	0	0		-		-		-
Glass	G1	Shaded			6 Btu/hr-sf	0	0		-		-		-
Door	D1	Door	21	0.50	10.50 Btu/hr-sf	0	0		-		-		-
Door	D2	Door	15	0.50	7.50 Btu/hr-sf	0	0		-		-		-
Above Grade Walls	W1	North	13	0.08	1.00 Btu/hr-sf	394	394	-	-	-	-	-	-
Above Grade Walls	W1	NorthEast	19	0.08	1.46 Btu/hr-sf	0	0	-	-	-	-	-	-
Above Grade Walls	W1	East	23	0.08	1.77 Btu/hr-sf	287	508	-	-	-	-	130	230
Above Grade Walls	W1	SouthEast	21	0.08	1.62 Btu/hr-sf	0	0	-	-	-	-	-	-
Above Grade Walls	W1	South	16	0.08	1.23 Btu/hr-sf	420	517	-	-	-	-	139	171
Above Grade Walls	W1	SouthWest	21	0.08	1.62 Btu/hr-sf	0	0	-	-	-	-	-	-
Above Grade Walls	W1	West	23	0.08	1.77 Btu/hr-sf	369	653	-	-	-	-	-	-
Above Grade Walls	W1	NorthWest	19	0.08	1.46 Btu/hr-sf	0	0	-	-	-	-	-	-
Basement Walls	W1	North	13	0.08	1.00 Btu/hr-sf	0	0	-	-	-	-	-	-
Basement Walls	W1	NorthEast	19	0.08	1.46 Btu/hr-sf	0	0	-	-	-	-	-	-
Basement Walls	W1	East	23	0.08	1.77 Btu/hr-sf	0	0	-	-	-	-	-	-
Basement Walls	W1	SouthEast	21	0.08	1.62 Btu/hr-sf	0	0	-	-	-	-	-	-
Basement Walls	W1	South	16	0.08	1.23 Btu/hr-sf	0	0	-	-	-	-	-	-
Basement Walls	W1	SouthWest	21	0.08	1.62 Btu/hr-sf	0	0	-	-	-	-	-	-
Basement Walls	W1	West	23	0.08	1.77 Btu/hr-sf	0	0	-	-	-	-	-	-
Basement Walls	W1	NorthWest	19	0.08	1.46 Btu/hr-sf	0	0	-	-	-	-	-	-
Other	W2	Grade	0	0.07	0.00 Btu/hr-sf	0	0	-	-	-	-	-	-
Other	W1	Partition	10	0.08	0.77 Btu/hr-sf	0	0	-	-	-	-	-	-
Other	W1	Garage	30	0.08	2.31 Btu/hr-sf	88	202	88	202	-	-	-	-
Other	W3	Attic	15	0.05	0.80 Btu/hr-sf	0	0	-	-	-	-	-	-
Other	W3	Kneewall	15	0.05	0.80 Btu/hr-sf	0	0	-	-	-	-	-	-
Roof	R1	Ceiling	55	0.03	1.71 Btu/hr-sf	1492	2544	45	77	94	160	397	677
Roof	R2	Vaulted	55	0.03	1.82 Btu/hr-sf	0	0		-		-		-
Floor	F1	Crawl/Uncon.	11	0.05	0.58 Btu/hr-sf	0	0		-		-		-
Floor	F1	Garage	30	0.05	1.59 Btu/hr-sf	369	587		-		-		-
Floor	F2	Slab	0	0.10	0.00 Btu/hr-sf	39	0	39	-		-		-
Internal	People				255 Btu/hr	4	1020		-		-		-
Internal	Lights (W)				3.413 Btu/hr-W	749	2556	23	78	47	160	199	679
Internal	Appliances (W)				3.413 Btu/hr-W	250	853	100	341	150	512		-
Vent	Infiltration		0.25	ACH	0.09 Btu/hr-cf		1249		77		76		322
Vent	Volume				CF	13878		855		846		3573	
Vent	Area				SF	1492		45		94		397	
Subtotal (Btu/hr)							13211		776		909		2498
Safety Factor				10 %			1321						
Sensible Cooling Load (Btu/hr)							**14533**						
Nominal Tons Required (75% sens. cap.)							**1.61**						
System Selection			2	Tons	Space CFM		**800**		**47**		**55**		**151**

Type	Description	Master Sitting Qty	Master Sitting Btu/hr	Master WIC (Left) Qty	Master WIC (Left) Btu/hr	Master WIC (Interior) Qty	Master WIC (Interior) Btu/hr	Master Bath Qty	Master Bath Btu/hr	Bed 2 Qty	Bed 2 Btu/hr
Ceiling Height		9		9		9		9		9	
G1	North		-		-		-		-		-
G1	NorthEast		-		-		-		-		-
G1	East	15	174		-		-		-		-
G1	SouthEast		-		-		-		-		-
G1	South		-		-		-		-		-
G1	SouthWest		-		-		-		-		-
G1	West		-		-		-	16	413	15	387
G1	NorthWest		-		-		-		-		-
G1	Sky Light		-		-		-		-		-
G1	Shaded		-		-		-		-		-
D1	Door		-		-		-		-		-
D2	Door		-		-		-		-		-
W1	North	68	68	-	-	-	-	-	-	-	-
W1	NorthEast	-	-	-	-	-	-	-	-	-	-
W1	East	74	131	-	-	-	-	-	-	-	-
W1	SouthEast	-	-	-	-	-	-	-	-	-	-
W1	South	68	84	73	90	-	-	139	171	-	-
W1	SouthWest	-	-	-	-	-	-	-	-	-	-
W1	West	-	-	-	-	-	-	142	252	103	183
W1	NorthWest	-	-	-	-	-	-	-	-	-	-
W1	North	-	-	-	-	-	-	-	-	-	-
W1	NorthEast	-	-	-	-	-	-	-	-	-	-
W1	East	-	-	-	-	-	-	-	-	-	-
W1	SouthEast	-	-	-	-	-	-	-	-	-	-
W1	South	-	-	-	-	-	-	-	-	-	-
W1	SouthWest	-	-	-	-	-	-	-	-	-	-
W1	West	-	-	-	-	-	-	-	-	-	-
W1	NorthWest	-	-	-	-	-	-	-	-	-	-
W2	Grade	-	-	-	-	-	-	-	-	-	-
W1	Partition	-	-	-	-	-	-	-	-	-	-
W1	Garage	-	-	-	-	-	-	-	-	-	-
W3	Attic	-	-	-	-	-	-	-	-	-	-
W3	Kneewall	-	-	-	-	-	-	-	-	-	-
R1	Ceiling	61	104	54	92	35	60	221	377	178	303
R2	Vaulted		-		-		-		-		-
F1	Crawl/Uncon.		-		-		-		-		-
F1	Garage	61	97	308	490		-		-		-
F2	Slab		-		-		-		-		-
People			-	1	255		-		-	1	255
Lights (W)		31	106	27	92	18	61	111	379	89	304
Appliances (W)			-		-		-		-		-
Infiltration			49		44		28		179		144
Volume		549		486		315		1989		1602	
Area		61		54		35		221		178	
Subtotal (Btu/hr)			814		1063		149		1771		1576
Safety Factor											
Sensible Cooling Load (Btu/hr)											
Nominal Tons Required (75% sens. cap.)											
System Selection			**49**		**64**		**9**		**107**		**95**

Type	Description	Bed 4 Qty	Bed 4 Btu/hr	Bath Qty	Bath Btu/hr	Bed 3 Qty	Bed 3 Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr
Ceiling Height		9		9		9		9		9		9	
G1	North		-		-		-		-		-		-
G1	NorthEast		-		-		-		-		-		-
G1	East		-		-	30	348		-		-		-
G1	SouthEast		-		-		-		-		-		-
G1	South		-		-		-		-		-		-
G1	SouthWest		-		-		-		-		-		-
G1	West	15	387		-		-		-		-		-
G1	NorthWest		-		-		-		-		-		-
G1	Sky Light		-		-		-		-		-		-
G1	Shaded		-		-		-		-		-		-
D1	Door		-		-		-		-		-		-
D2	Door		-		-		-		-		-		-
W1	North	135	135	53	53	138	138	-	-	-	-	-	-
W1	NorthEast	-	-	-	-	-	-	-	-	-	-	-	-
W1	East	-	-	-	-	83	146	-	-	-	-	-	-
W1	SouthEast	-	-	-	-	-	-	-	-	-	-	-	-
W1	South	-	-	-	-	-	-	-	-	-	-	-	-
W1	SouthWest	-	-	-	-	-	-	-	-	-	-	-	-
W1	West	123	218	-	-	-	-	-	-	-	-	-	-
W1	NorthWest	-	-	-	-	-	-	-	-	-	-	-	-
W1	North	-	-	-	-	-	-	-	-	-	-	-	-
W1	NorthEast	-	-	-	-	-	-	-	-	-	-	-	-
W1	East	-	-	-	-	-	-	-	-	-	-	-	-
W1	SouthEast	-	-	-	-	-	-	-	-	-	-	-	-
W1	South	-	-	-	-	-	-	-	-	-	-	-	-
W1	SouthWest	-	-	-	-	-	-	-	-	-	-	-	-
W1	West	-	-	-	-	-	-	-	-	-	-	-	-
W1	NorthWest	-	-	-	-	-	-	-	-	-	-	-	-
W2	Grade	-	-	-	-	-	-	-	-	-	-	-	-
W1	Partition	-	-	-	-	-	-	-	-	-	-	-	-
W1	Garage	-	-	-	-	-	-	-	-	-	-	-	-
W3	Attic	-	-	-	-	-	-	-	-	-	-	-	-
W3	Kneewall	-	-	-	-	-	-	-	-	-	-	-	-
R1	Ceiling	190	324	60	102	157	268		-		-		-
R2	Vaulted		-		-		-		-		-		-
F1	Crawl/Uncon.		-		-		-		-		-		-
F1	Garage		-		-		-		-		-		-
F2	Slab		-		-		-		-		-		-
People		1	255		-	1	255		-		-		-
Lights (W)		95	324	30	102	79	270	-	-	-	-	-	-
Appliances (W)			-		-		-		-		-		-
Infiltration			154		49		127		-		-		-
Volume		1710		540		1413		0		0		0	
Area		190		60		157							
Subtotal (Btu/hr)			1798		306		1552		0		0		0
Safety Factor													
Sensible Cooling Load (Btu/hr)													
Nominal Tons Required (75% sens. cap.)													
System Selection			**109**		**19**		**94**		**-**		**-**		**-**

HEATING LOAD CALCULATION

Components	Type	Description	ΔT (°F)	U-Value	Heating Load Factor	Entire House Qty	Entire House Btu/hr	Stairs Qty	Stairs Btu/hr	Landing Qty	Landing Btu/hr	Master Bed Qty	Master Bed Btu/hr
Glass	G1		54	0.33	17.8 Btu/hr-sf	121	2156	0	-	0	-	30	535
Glass	G2		54	0.00	0.0 Btu/hr-sf	0	0	0	-	0	-	0	-
Glass	G3		54	0.00	0.0 Btu/hr-sf	0	0	0	-	0	-	0	-
Door	D1		54	0.50	27.0 Btu/hr-sf	0	0	0	-	0	-	0	-
Door	D2		54	0.50	27.0 Btu/hr-sf	0	0	0	-	0	-	0	-
Walls	W1		54	0.08	4.2 Btu/hr-sf	1470	6111	0	-	0	-	269.2	1,119
Walls	W2		54	0.07	3.6 Btu/hr-sf	0	0	0	-	0	-	0	-
Walls	W3		54	0.05	2.9 Btu/hr-sf	0	0	0	-	0	-	0	-
Walls	W1	Partition	27	0.08	2.1 Btu/hr-sf	88	182	87.63	182	0	-	0	-
Walls	W2	Grade	27	0.07	1.8 Btu/hr-sf	0	0	0	-	0	-	0	-
Roof	R1		54	0.03	1.7 Btu/hr-sf	1492	2498	45	75	94	157	397	665
Roof	R2		54	0.03	1.8 Btu/hr-sf	0	0	0	-	0	-	0	-
Floor	F1		54	0.05	2.9 Btu/hr-sf	369	1056	0	-	0	-	0	-
Floor	F2	Slab	27	0.10	2.7 Btu/hr-sf	39	105	39	105	0	-	0	-
Area					SF	1492		45		94		397	
Infiltration			0.25	ACH	0.24 Btu/hr	0	3372		208		206		868
Subtotal (Btu/hr)							15481		571		363		3187
Safety Factor				15 %			2322						
Heat Loss (Btu/hr)							**17803**						
Minimum Heater Size (kW)							**5.22**						
System Selection			8	kW	Htg. CFM		**800**		**29**		**19**		**165**

Type	Description	Master Sitting Qty	Master Sitting Btu/hr	Master WIC Qty	Master WIC Btu/hr	Master WIC Qty	Master WIC Btu/hr	Master Bath Qty	Master Bath Btu/hr	Bed 2 Qty	Bed 2 Btu/hr
G1		15	267	0	-	0	-	16	285	15	267
G2		0	-	0	-	0	-	0	-	0	-
G3		0	-	0	-	0	-	0	-	0	-
D1		0	-	0	-	0	-	0	-	0	-
D2		0	-	0	-	0	-	0	-	0	-
W1		210.8	877	73.3	305	0	-	281.5	1,170	103.3	430
W2		0	-	0	-	0	-	0	-	0	-
W3		0	-	0	-	0	-	0	-	0	-
W1	Partition	0	-	0	-	0	-	0	-	0	-
W2	Grade	0	-	0	-	0	-	0	-	0	-
R1		61	102	54	90	35	59	221	370	178	298
R2		0	-	0	-	0	-	0	-	0	-
F1		61	175	308	881	0	-	0	-	0	-
F2	Slab	0	-	0	-	0	-	0	-	0	-
Area		61		54		35		221		178	
Infiltration			133		118		77		483		389
Subtotal (Btu/hr)			1554		1395		135		2309		1384
Safety Factor											
Heat Loss (Btu/hr)											
Minimum Heater Size (kW)											
System Selection			**80**		**72**		**7**		**119**		**72**

Type	Description	Bed 4 Qty	Bed 4 Btu/hr	Bath Qty	Bath Btu/hr	Bed 3 Qty	Bed 3 Btu/hr	0 Qty	0 Btu/hr	0 Qty	0 Btu/hr	0 Qty	0 Btu/hr
G1		15	267	0	-	30	535	0	-	0	-	0	-
G2		0	-	0	-	0	-	0	-	0	-	0	-
G3		0	-	0	-	0	-	0	-	0	-	0	-
D1		0	-	0	-	0	-	0	-	0	-	0	-
D2		0	-	0	-	0	-	0	-	0	-	0	-
W1		258.3	1,074	52.5	218	220.8	918	0	-	0	-	0	-
W2		0	-	0	-	0	-	0	-	0	-	0	-
W3		0	-	0	-	0	-	0	-	0	-	0	-
W1	Partition	0	-	0	-	0	-	0	-	0	-	0	-
W2	Grade	0	-	0	-	0	-	0	-	0	-	0	-
R1		190	318	60	100	157	263	0	-	0	-	0	-
R2		0	-	0	-	0	-	0	-	0	-	0	-
F1		0	-	0	-	0	-	0	-	0	-	0	-
F2	Slab	0	-	0	-	0	-	0	-	0	-	0	-
Area		190		60		157		0		0		0	
Infiltration			416		131		343		-		-		-
Subtotal (Btu/hr)			2075		450		2059		0		0		0
Safety Factor													
Heat Loss (Btu/hr)													
Minimum Heater Size (kW)													
System Selection			**107**		**23**		**106**		**-**		**-**		**-**